

Multiple Choice (10 marks)

1. During a semester, a student received scores of 76, 80, 83, 71, 80, and 78 on six tests. What is the student's average score for these six tests?
 - (a) 76
 - (b) 77
 - (c) 78
 - (d) 79
2. Solve $18.5 \text{ dol over m gal} = 3.60 \text{ dol over } 7.5 \text{ gal}$. Round to the nearest hundredth if necessary.
 - (a) 8.86
 - (b) 42.25
 - (c) 32.54
 - (d) 38.44
3. What is the smallest positive integer with factors of 16, 15, and 12?
 - (a) 12
 - (b) 22
 - (c) 840
 - (d) 240
4. What is the shortest distance from the origin to the circle defined by $x^2 - 24x + y^2 + 10y + 160 = 0$?
 - (a) 10
 - (b) 16
 - (c) 24
 - (d) 12
5. Evaluate $-2(x - 3)$ for $x = 2$.
 - (a) -4
 - (b) -2
 - (c) 10
 - (d) 2

True / False (10 marks)

Answer True or False

6. True or False: Jane's mean quiz score is 92, which is the lowest score among her quiz results.
7. True or False: The ratio of x to y is 4 to 5 when the ratio of $2x - y$ to $x + y$ is 2 to 3.
8. True or False: The interval $[-1, 1]$ consists of all values of u such that neither $2u$ nor $-20u$ is in the interval $(-, -1)$.
9. True or False: The correct answer to 'Evaluate $\log_3 81$.' is '0.25'.
10. True or False: You can buy 36 cans of pears for 9.48 if 3 cans cost 2.37.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. An ice cream shop sold 48 vanilla milkshakes, which accounted for 40% of the total milkshakes sold that day. Calculate the total number of milkshakes sold and explain your reasoning.
12. A baker made 232 muffins and sent 190 to a hotel. Explain how to determine the number of boxes needed for the remaining muffins, including the equation involved in your solution.

End of Paper